C++ Notes: Building a Better Calculator

(4 operations: +, -, /, *)

Author: Ba Tung Le

Learning Objectives

After this section, you will be able to:

- Read two numbers and an operator from the user
- Use `if / else if / else` to choose an operation
- Handle invalid operators safely
- Avoid printing an uninitialized result

Core Idea

Core idea

We input:

- `num1`
- op (one character: `+ - * /`)
- `num2`

Then we choose the correct calculation using `if` statements.

Full Program (recommended: use double)

Listing 1: Four-function calculator

```
#include <iostream>
using namespace std;

int main() {
  double num1, num2;
  char op;

  cout << "Enter first number: ";
  cin >> num1;

  cout << "Enter operator (+, -, *, /): ";
  cin >> op;

  cout << "Enter second number: ";
```

```
  cin >> num2;

  double result = 0.0;
  bool valid = true;

  if (op == '+') {
    result = num1 + num2;
  } else if (op == '-') {
    result = num1 - num2;
  } else if (op == '/') {
    if (num2 == 0) {
      cout << "Error: division by zero" << endl;
      valid = false;
    } else {
      result = num1 / num2;
    }
  } else if (op == '*') {
    result = num1 * num2;
  } else {
    cout << "Invalid operator" << endl;
    valid = false;
  }

  if (valid) {
    cout << result << endl;
  }

  return 0;
}
```

Example Runs

Addition

Input: `5 + 30`

Output:

```
Enter first number: 5
Enter operator (+, -, *, /): +
Enter second number: 30
35
```

Division

Input: `50 / 5`

Output:

```
Enter first number: 50
Enter operator (+, -, *, /): /
```

```
Enter second number: 5
10
```

Invalid operator

Input: `5 G 9`

Output:

```
Enter first number: 5
Enter operator (+, -, *, /): G
Enter second number: 9
Invalid operator
```

Why we use `valid` and initialize `result`

If the operator is invalid, we should **not** print `result`. Otherwise, you may print garbage (uninitialized value).

Ba Tung Le

Email: tombn2607@gmail.com